VoteChain

Blockchain-Based Voting System



Developed and Presented by

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GitHub Repository:

https://github.com/goswamiumang108/Blockchain_Based_Voting_System

THE PROBLEM

Online elections need stronger trust

Common risks in small-scale college and club elections

01 **Centralized records**

A single database or spreadsheet becomes the point of trust.

02 **Silent changes**

Votes may be edited without a visible audit trail.

03 **Duplicate voting**

Interface-only checks can fail to enforce one vote.

04 **Ballot exposure**

Linking names to choices threatens voter secrecy.

Five goals. One verifiable workflow.

01	**Authenticate**	Hashed passwords for registration and login
02	**Enforce**	One vote per eligible voter
03	**Record**	Custom blockchain with Proof of Work
04	**Separate**	Keep voter identity apart from ballot data
05	**Validate**	Count results only from validated votes

From registration to verified results

01 Register

Create an account; admin enrolls voters

02 Open election

Admin opens the voting window

03 Vote

One ballot per eligible voter

04 Mine

Seal the ballot into a block

05 Verify

Check the whole chain

06 Results

Count validated votes

Hash links make changes detectable

Each block stores its own hash and the previous block’s hash.

Block 0 / Genesis
HASH + PREVIOUS HASH

→

Block 1
HASH + PREVIOUS HASH

→

Block 2
HASH + PREVIOUS HASH

FIELDS index · timestamp · transactions · previous hash · nonce · hash

Proof of Work: search for a nonce until SHA-256 starts with “000”.

Defense at multiple layers

Passwords	bcrypt hashes; no plaintext storage
One vote	Server checks and SQLite constraints
Ballot privacy	Random receipt ID and keyed voter tag; no name in the ballot
Integrity	Validate the chain before counting results

A lightweight stack. A visible audit trail.

Python / Streamlit / SQLite / hashlib · SHA-256 / bcrypt

Vote page

SCREENSHOT PLACEHOLDER

Show ballot selection and vote confirmation.

Results page

SCREENSHOT PLACEHOLDER

Show totals from validated votes.

Blockchain Explorer

SCREENSHOT PLACEHOLDER

Show the chain status: “Chain verified”.

Replace these placeholders with screenshots from the working app.

A foundation for verifiable voting

PROJECT OUTCOMES

Voting application

Tamper-evident blockchain

Results from validated votes

NEXT STEPS

Multi-node deployment

Stronger ballot privacy

Explore zero-knowledge proofs

Thank you